ENGINEERING TRAINING MODULE · USB-C / PLATFORM POWER · REV A

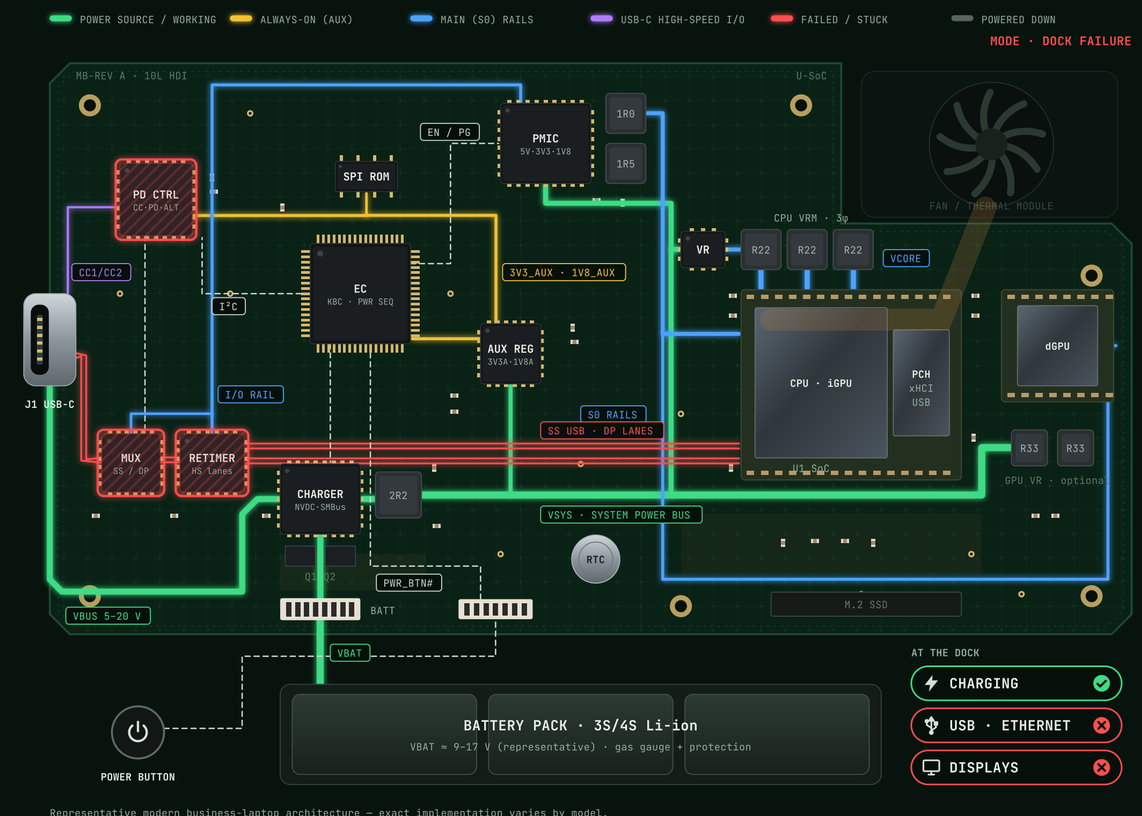
USB-C DOCK FAILURE EXPLAINED

HP USB-C Dock G5 → Windows business laptop · charging kept working · USB, Ethernet, displays gone — dock absent from Device Manager · restart did not help · ≈45 s hold fixed it

Representative architecture — exact implementation varies by model. The successful reset does not prove which controller failed.



01 PCB MOTHERBOARD · WHERE THE STATE LIVES

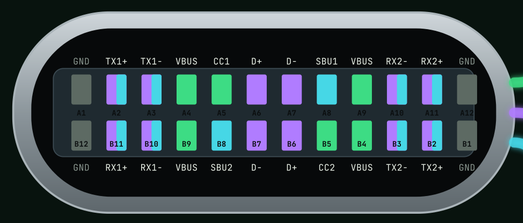


Representative modern business-laptop architecture — exact implementation varies by model.

02 USB-C SIGNAL PATH · ONE CONNECTOR, THREE JOBS

ONE CONNECTOR ≠ ONE SUBSYSTEM

USB-C receptacle, viewed face-on · 24 contacts · three independent jobs



- POWER · VBUS + CC (PD messages ride CC)
- USB 2.0 · D+/D-
- SUPERSPEED PAIRS — shared: USB 3 or DisplayPort, set by the mux
- SBU1/2 → DP AUX in Alt Mode
- GROUND

Plug is reversible: CC tells the controller which way round it is, then the mux maps TX1/RX1 or TX2/RX2 to the right destination.

POWER
VBUS · 5 / 9 / 15 / 20 V
CC1 / CC2 · attach + orientation
USB Power Delivery contract

USB DATA
D+ / D- · USB 2.0
SuperSpeed TX / RX pairs · USB 3
Dock hub → USB ports + Ethernet

DISPLAYPORT
High-speed lanes (2 or 4, via mux)
SBU1 / SBU2 → DP AUX channel
DisplayPort Alt Mode

INCIDENT
POWER — WORKING
USB DATA — MISSING
DISPLAYPORT — MISSING

Charging proves VBUS + CC + PD are healthy.
It proves nothing about the data and display paths.

Power finished early and stayed up. Not even USB 2.0 enumerated, so the stuck step sits before data AND display: the USB connection / data-role state the PD controller reports.

03 VOLTAGE DECAY · WHEN STATE IS ACTUALLY LOST



$V(t) = V_0 \cdot e^{(-t/RC)}$ · state survives until the rail crosses the retention threshold

04 WARM VS HARDWARE RESET

WINDOWS RESTART

warm reset · OS-controlled

CPU → RESET — core reset
WINDOWS → RELOAD — drivers re-init

BUT

AUX POWER RAIL → STILL ON — never switched
USB-C CONTROLLER → POWERED — keeps its RAM
STUCK STATE → SURVIVES — same bad state

RESULT: POWER ✓ · USB ✗ · DISPLAY ✗

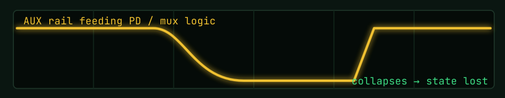


HARDWARE POWER-STATE RESET

long power-button hold · EC-controlled

EC / POWER SEQUENCER — reads PWR_BTN#
REGULATORS DISABLED — EN pins low
RELEVANT RAILS COLLAPSE — caps discharge
CONTROLLER LOSES STATE — volatile cleared
COLD INITIALISATION — firmware reloads
USB-C NEGOTIATES AGAIN — CC → PD → Alt Mode

RESULT: POWER ✓ · USB ✓ · DISPLAY ✓



05 RESET TIMELINE · IS 45 s SPECIAL?



No verified procedure needs 45 s; all documented thresholds are ≤ 40 s. 45 s = margin, not magic.

06 USB-C REINITIALISATION · CC → PD → ALT → USB → OS

NORMAL USB-C DOCK BRING-UP

Representative order · USB 2.0 can enumerate as soon as VBUS is up; DP and USB 3 wait for Alt Mode

1. CABLE INSERTED — plug mates · orientation unknown
2. CC ATTACH — Rp/Rd seen on CC1 or CC2 → flip known
3. PD NEGOTIATION — Source_Caps → Request → Accept → PS_RDY
4. VBUS ESTABLISHED — contract voltage, e.g. 20 V
5. CHARGING — charger takes VBUS → VSYS + battery
6. DP ALT MODE — Discover ID/SVIDs/Modes → Enter Mode
7. MUX ROUTES LANES — pin assignment C or D · orientation
8. USB HUB ENUMERATES — dock hub, NIC, audio, ports
9. DP LINK TRAINS — AUX over SBU · clock recovery · EQ
10. WINDOWS ENUMERATES — PnP: displays, Ethernet, USB devices

POWER DOMAIN · stages 1–5
DATA / ALT MODE · 6–8
DISPLAY · 9
OS · 10

CHARGING ✓ · USB ✓ · DISPLAY ✓

After a clean reset, the chain re-runs from stage 1 and every function returns.

07 DIAGNOSTIC DECISION PATH · THIS INCIDENT

DOCK CONNECTED → CHARGING? YES → USB WORKING? NO → DISPLAYS? MISSING → IN DEVICE MGR? NOT AT ALL → PRE-USB STATE PD conn · EC → RESTART FIXED? NO → HARD RESET hold power → RETURNED? YES → TRANSIENT STATE strong hypothesis

Side exits: not charging → charger / cable / PD · USB OK → display path only · dock visible in Device Manager → Alt Mode / lane path · restart fixes it → driver / OS state · reset fails → hardware fault

Sources: ACPI 6.5 §4.8.2.2.1.3 · HP Support "Perform a power reset" · HP USB-C Dock G5 datasheet + SoftPaq firmware notes · Dell KB 000139016, 000125880 · Lenovo HT510479 · Microsoft Support 4036280